IG_1/IG_1-FP/RT_1

Examen

06 Juillet 2021

Durée: 2^h

Calcul Intégral

Exercice I

Calculer les primitives suivantes :

1) $\displaystyle\int \frac{x^2}{x^3+1}dx.$

2) $\displaystyle\int \cos(x)\sin(x)dx.$

3) $\displaystyle\int \sqrt{1-x}dx.$

4) $\displaystyle\int x\sqrt{1+x^2}dx.$

Exercice II

A l'aide d'une intégration par partie calculer les intégrales suivantes :

1) $I_1 = \displaystyle\int_0^{\frac{\pi}{2}} e^t\cos(t)dt$

2) $I_2 = \displaystyle\int_0^{\frac{\pi}{2}} t\sin^3(t)dt$

3) $I_3 = \displaystyle\int_0^{\pi} (t-1)\cos(t)dt$

Exercice III

1) Décomposer en éléments simple

$$f(x) = \frac{2x+1}{x^2(x+1)}.$$

2) Calculer

$$F(t) = \int_1^t \frac{1}{x^2}\ln(x^2+x)dx.$$

Exercice IV

Calculer les intégrales suivantes en utilisant le changement de variable précisé :

1) $A = \displaystyle\int_1^e \frac{\ln(t)}{t}dt$, en posant $u = \ln t$.

2) $B = \displaystyle\int_0^{\frac{\pi}{2}} \sin(t)\cos^2(t)dt$, en posant $u = \cos(t)$.

3) $C = \displaystyle\int_0^{\frac{\pi}{4}} (\tan(t))^4dt$, en posant $u = \tan(t)$.

4) $D = \displaystyle\int_1^e \frac{\ln(t)}{\sqrt{t}}dt$, en posant $u = \sqrt{t}$.

Examen 2021

Exo 1:

① $\int \frac{x^2}{x^3+1}\,dx = \int \frac{1}{3}\times\frac{3x^2}{x^3+1}\,dx$

$= \frac{1}{3}\ln(x^3+1) + cte$

② $\int \cos(x)\sin(x)\,dx$

$= \frac{1}{2}\sin^2(x) + cte$

③ $\int \sqrt{1-x}\,dx = \int (1-x)^{\frac{1}{2}}\,dx$

$= -\int -(1-x)^{\frac{1}{2}}\,dx$

$= -\frac{2}{3}(1-x)^{\frac{3}{2}} + cte$

④ $\int x\sqrt{1+x^2}\,dx = \int x(1+x^2)^{\frac{1}{2}}\,dx$

$= \frac{1}{2}\int 2x(1+x^2)^{\frac{1}{2}}\,dx$

$= \frac{1}{2}\times\frac{2}{3}(1+x^2)^{\frac{3}{2}} + cte$

$= \frac{1}{3}(1+x^2)^{\frac{3}{2}} + cte$

Exo 2:

① $I_1 = \int_0^{\frac{\pi}{2}} e^t\cos(t)\,dt$

D'après I.I.P

$U(t) = e^t \qquad V'(t) = \cos t$

$U'(x) = e^t \qquad V(t) = \sin t$

$\int_0^{\frac{\pi}{2}} e^t\cos(t)\,dt = \left[e^t\sin t\right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} e^t\sin t\,dt$

on fait une deuxième I.I.P

$U(t) = e^t \qquad V'(t) = \sin t$

$U'(t) = e^t \qquad V(t) = -\cos t$

$\int_0^{\frac{\pi}{2}} e^t\sin t\,dt = \left[-e^t\cos t\right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} -\cos t\, e^t\,dt$

$= -\left[e^t\cos t\right]_0^{\frac{\pi}{2}} + \int_0^{\frac{\pi}{2}} e^t\cos t\,dt.$

donc

$I_1 = \int_0^{\frac{\pi}{2}} e^t\cos t\,dt = \left[e^t\sin t\right]_0^{\frac{\pi}{2}} + \left[e^t\cos t\right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} e^t\cos t\,dt$

$I_1 = 2\int_0^{\frac{\pi}{2}} e^t\cos t\,dt = \left[e^t\sin t\right]_0^{\frac{\pi}{2}} + \left[e^t\cos t\right]_0^{\frac{\pi}{2}}$

$I_1 = \int_0^{\frac{\pi}{2}} e^t\cos t\,dt = \frac{1}{2}\left[e^t\sin t + e^t\cos t\right]_0^{\frac{\pi}{2}}$

$= \frac{e^{\frac{\pi}{2}}}{2}$

② $I_2 = \int_0^{\frac{\pi}{2}} t\sin^3(t)\,dt$

Ex 02:

$$\int_0^{\frac{\pi}{2}} t\sin^3(t)\,dt$$

I.P.P

$U(t)=t \qquad V'(t)=\sin^3 t$

$U'(t)=1 \qquad V(t)=-\cos t+\frac{\cos^3 t}{3}$

$$\int_0^{\frac{\pi}{2}} t\sin^3 t\,dt=\left[t\left(-\cos t+\frac{\cos^3 t}{3}\right)\right]_0^{\frac{\pi}{2}}-\int_0^{\frac{\pi}{2}}\left(-\cos t+\frac{\cos^3 t}{3}\right)dt$$

$$=\left[t\left(-\cos t+\frac{\cos^3 t}{3}\right)\right]_0^{\frac{\pi}{2}}$$

$$-\int_0^{\frac{\pi}{2}}\cos t+\frac{\cos t(1-\sin^2 t)}{3}\,dt$$

$$=\left[t\left(-\cos t+\frac{\cos^3 t}{3}\right)\right]_0^{\frac{\pi}{2}}$$

$$-\left[\sin t+\frac{\sin t}{3}-\frac{1}{3}\cdot\frac{1}{3}\sin^3 t\right]_0^{\frac{\pi}{2}}$$

$$1-\frac{1}{3}+\frac{1}{9}=\frac{9-3+1}{9}$$

$$=\frac{7}{9}$$

(3) $I_3=\int_0^{\pi}(t-1)\cos t\,dt$

I.P.P

$U(t)=t-1 \qquad V'(t)=\cos t$

$U'(t)=1 \qquad V(t)=\sin t$

$$\int_0^{\pi}(t-1)\cos t\,dt=\left[(t-1)\sin t\right]_0^{\pi}-\int_0^{\pi}\sin t\,dt$$

$$=\left[(t-1)\sin t\right]_0^{\pi}-\left[-\cos t\right]_0^{\pi}$$

$$=-1-1=-2$$

Exo3:

1)

$$\frac{2n+1}{n^2(n+1)}$$

$$\frac{2n+1}{n^2(n+1)} = \frac{a}{n} + \frac{b}{n^2} + \frac{c}{(n+1)} \quad (*)$$

* on multiple (*) par $(n+1)$ on prend $n=-1$.

$$\frac{(2n+1)\cancel{(n+1)}}{n^2\cancel{(n+1)}} = \frac{a(n+1)}{n} + \frac{b(n+1)}{n^2} + \frac{c\cancel{(n+1)}}{\cancel{(n+1)}}$$

on prend $n=-1$

$$\frac{-2+1}{1} = \frac{a(-1+1)}{n} + \frac{b(-1+1)}{n^2} + c$$

$\Rightarrow c = -\frac{1}{1} = -1$ $\quad \boxed{c=-1}$

* on multiple (*) par n on fait tendre $n \to +\infty$

$$\lim_{+\infty}\left(\frac{2n^2+n}{n^2(n+1)}\right) = \lim_{n\to+\infty}\left(\frac{an}{n} + \frac{bn}{n^2} + \frac{cn}{n+1}\right)$$

$0 = a + c \Rightarrow a = -c$ $\quad \boxed{a=1}$,

* on calcule $f(1)$.

$$\frac{2+1}{2} = a + b + \frac{c}{2} = 1 + b - \frac{1}{2}$$

$\Rightarrow b = \frac{3}{2} + \frac{1}{2} - 1 = \frac{2}{2} = 1$

$\boxed{b=1}$

$$\boxed{\frac{2n+1}{n^2(n+1)} = \frac{1}{n} + \frac{1}{n^2} - \frac{1}{n+1}}$$

2) $F(t) = \int_1^t \frac{1}{n^2}\ln(n^2+n)\,dn$

D'aprel I.P.P

$U(n) = \ln(n^2+n) \quad V'(n) = \frac{1}{n^2}$

$U'(n) = \frac{2n+1}{n^2+n} \quad V(n) = -\frac{1}{n}$

$$\int_1^t \frac{1}{n^2}\ln(n^2+n)\,dn = \left[-\frac{1}{n}\ln(n^2+n)\right]_1^t + \int_1^t \frac{2n+1}{n^2(n+1)}\,dn$$

donc

$$F(t) = \left[-\frac{1}{n}\ln(n^2+n)\right]_1^t + \int_1^t \left(\frac{1}{n} + \frac{1}{n^2} - \frac{1}{n+1}\right)dn$$

$$= \left[-\frac{1}{n}\ln(n^2+n)\right]_1^t + \left[\ln n - \frac{1}{n} - \ln(n+1)\right]_1^t$$

$$= -\frac{1}{t}\ln(t^2+t) + \ln t - \frac{1}{t} - \ln(t+1) + \ln(2) - 0 + 1 + \ln 2$$

$$= -\frac{1}{t}\ln(t^2+t) + \ln t - \frac{1}{t} - \ln(t+1) + 2\ln 2 + 1$$

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... .

① $A = \int_1^e \frac{\ln(t)}{t} dt$

on pose

$U = \ln(t)$.

$dU = \frac{1}{t} dt \Rightarrow \boxed{dt = t\, dU}$

$t \to 1 \quad U \to 0$

$t \to e \quad U \to 1$

$A = \int_1^e \frac{\ln(t)}{t} dt = \int_0^1 \frac{U}{t} \times t\, dU$

$= \int_0^1 U\, dU = \left[\frac{1}{2} U^2\right]_0^1$

$= \frac{1}{2} - 0 = \frac{1}{2}$

② $B = \int_0^{\frac{\pi}{2}} \sin t \cos^2(t)\, dt$

on pose $U = \cos(t)$

$dU = -\sin t\, dt$

$dt = -\frac{dU}{\sin t}$

$t \to 0 \quad U \to 1$

$t \to \frac{\pi}{2} \quad U \to 0$

$B = \int_0^{\frac{\pi}{2}} \sin(t) \cos^2(t)\, dt = \int_1^0 -\sin t\, U^2 \times \frac{dU}{\sin(t)}$

$= \int_0^1 U^2\, dU = \left[\frac{1}{3} U^3\right]_0^1 = \frac{1}{3}$

③ CF $\int_0^{\frac{\pi}{4}} (\tan(t))^4\, dt$

on pose ~~U = tan~~

$U = \tan(t)$

$dU = (1 + \tan^2(t))\, dt$

$\boxed{dt = \frac{dU}{1 + \tan^2 t}}$

$t \to 0 \quad U \to 0$

$t \to \frac{\pi}{4} \quad U \to 1$

$\int_0^{\frac{\pi}{4}} (\tan(t))^4\, dt = \int_0^1 U^4 \times \frac{dU}{1 + \tan^2 t}$

$= \int_0^1 \frac{U^4}{1 + U^2}\, dt = \int_0 \frac{U^4 + U^2 - U^2}{1 + U^2}\, dU$

$= \int_0^1 \frac{U^2(U^2 + 1)}{1 + U^2} - \frac{U^2}{1 + U^2}\, dU$

$= \int_0^1 U^2 - \frac{(U^2 + 1 - 1)}{1 + U^2}\, dU$

$= \int_0^1 U^2 - 1 + \frac{1}{1 + U^2}\, dU$

$= \left[\frac{1}{3} U^3 - U + \arctan(U)\right]_0^1$

$= \frac{1}{3} - 1 + \frac{\pi}{4} - 0$

$= \frac{\pi}{4} - \frac{2}{3}$

4/

$$\int_1^e \frac{\ln(t)}{\sqrt{t}}\,dt$$

on pose $U=\sqrt{t}$

$$dU = \frac{1}{2\sqrt{t}}\,dt$$

$$\boxed{dt = 2\sqrt{t}\,dU}$$

$t \to 1$ $\quad U \to 1$

$t \to e$ $\quad U \to \sqrt{e}$

$$\int_1^e \frac{\ln t}{\sqrt{t}}\,dt = \int_1^{\sqrt{e}} \frac{\ln(U^2)}{\sqrt{t}}\,2\sqrt{t}\,dU$$

$$= \int_1^{\sqrt{e}} 2\ln(U^2)\,dU$$

$$= 2\int_1^{\sqrt{e}} 2\ln(U)\,dU$$

$$= 4\left[U\ln U - U\right]_1^{\sqrt{e}}$$

$$= 4(\sqrt{e}\ln(\sqrt{e}) - \sqrt{e} + 4$$